

C01-AT-1

1) Solve the water jug problem:-

⇒ Two jugs are given:- one 4-gallon jug and one 3-gallon jug. There are no measuring marks on the jugs.

⇒ Water can be filled from a pump, emptied onto the ground, or poured from one jug to another

⇒ The objective is to obtain exactly 2 gallon of water in the 4-gallon jug.

Initial state:-

4-gallon jug = 0 gallons

3-gallon jug = 0 gallons

Goal state:-

4-gallon jug = 2 gallons

3-gallon jug = 0 gallons

Step:-

Action	4-gallon jug	3-gallon jug
1) fill the 3-gallon jug	0	3
2) Pour into the 4-gallon jug	3	0
3) Fill the 3-gallon jug again	3	3
4) Pour into 4-gallon jug until it is full	4	2
5) Empty the 4-gallon jug	0	2
6) Pour the remaining 2-gallon into the 4-gallon jug	2	0

Conclusion:-

The required 2 gallon is successfully obtained in 4-gallon jug.

2) Mars Rover Agent

A Mars Rover is an intelligent robotic agent that explores the surface of Mars. It collects samples, analyzes rock and soil, and sends information back to earth.

9) percepts.

camera image

Rock and soil sample

Temperature sensor

obstacle detector

battery level sensor

Position and navigation sensors.

ii) Operating Environment.

Partially observable environment

Dynamic environment

Sequential environment

continuous environment

Stochastic (uncertain) environment

Single-agent environment.

iii) Actions:-

Move forward or backward
Turn left or right
capture photographs
Drill rocks
Collect soil samples
Analyze collected sample
Avoid obstacles
Recharge using solar panels
Send scientific data to earth.

iv) performance Measure

Number of successful sample collected
Accuracy of scientific analysis
Distance explored
Battery efficiency
Safe navigation
Successful transmission of data

v) Suitable Agent Architecture

Utility Based agent.

A utility-based agent selects the best action by considering several factors such as battery power, safety, scientific importance, and mission objectives.

Since the mars environment is uncertain this agent provides better decision-making than simple or goal-based agents.

conclusion:-

A Utility-Based Agent is the most suitable architecture for a mars Rover.

3. 8-Queens problem:-

Problem Statement:-

The objective of 8-Queens problem is to place eight queens on 8×8 chessboard so that no queen attacks another queen.

A queen attacks another queen if they are in the same:-

Row
column
Diagonal.

Problem Components.

Initial State:-

An empty chessboard or a board with partially placed queens

State Space.

All possible arrangements of eight queens on the chessboard.

Actions:-

Place a queen in a safe position

Move a queen to another position

Backtrack when conflicts occur.

Goal test:-

The solution is achieved when:-

All eight queens are placed
No two queens attack each other.

Path cost.

Each move has a uniform cost of 1

Search Strategy

The most suitable search strategy is
Back tracking search.

Steps: -

⇒ Place the first queen in first column

⇒ Move to next column

⇒ Check whether the new queen attacks any previous queen

⇒ If there is conflict, back track

⇒ Continue until all eight queen are safely placed.

Conclusion

Backtracking efficiently find a valid arrangement where no queen attacks another queen.

Q. OLA Cab Booking Application.

Problem Statement.

A customer wants to travel from one location to another using the OLA cab application the customer selects the preferred cab type such as mini, micro, sedan, prime or shared.

Suitable - Problem - Solving Agent .

Goal - Based Agent .

The goal to the customer is to reach the destination using a preferred cab. The agent searches for available cabs, compares options, and books the cabs that satisfies the customer goal.

Pseudo Code:-

START

Enter pickup location

Enter destination

Select preferred cab type

Search available cabs

IF preferred cab is available THEN

Display fare

Confirm booking

Assign driver

ELSE

Display alternative cab types

END IF

Show driver details

Start trip

END

Conclusion:-

A Goal - Based Agent is suitable because it focuses on achieving the customer's goal of reaching the destination.

5. Uniform Cost Search (UCS)

Problem Statement:-

A logistics company wants to transport packages from warehouse S to warehouse G at minimum travel cost using the uniform cost search (UCS) algorithm.

Given Costs:-

$$\begin{array}{llll}
 S \rightarrow A = 1 & S \rightarrow G = 12 & A \rightarrow B = 3 & A \rightarrow C = 1 \\
 B \rightarrow D = 3 & C \rightarrow D = 3 & C \rightarrow D = 1 & C \rightarrow G = 2 \\
 & & D \rightarrow G = 3 &
 \end{array}$$

UCS Expansion

Step	Expanded Node	Total
1	S	0
2	A	1
3	C	2
4	D	3
5	G	4

Path cost calculation

Path	Cost
$S \rightarrow G$	12
$S \rightarrow A \rightarrow B \rightarrow D \rightarrow G$	10
$S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$	6
$S \rightarrow A \rightarrow C \rightarrow G$	4

optimal path.

$S \rightarrow A \rightarrow C \rightarrow G$

Minimum cost

4.

UCS Algorithm:-

STAR

Insert the start node into the priority queue.

WHILE the queue is not empty

Remove the node with the lowest cost.

IF the node is the goal THEN

Return the path and total cost.

ENDIF

Expand the node

Insert all child nodes with cumulative

END WHILE

STOP

Conclusion:-

Uniform Cost search always expands the node with the lowest cumulative cost and finds the optimal path. Therefore, the least-cost path is $S \rightarrow A \rightarrow C \rightarrow G$

Total cost = 4.